

TED UNIVERSITY

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Senior Design Project I

Project Specifications Report

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1. Introduction

1.1 Description

TRAQS is a web-based tool for managing and tracking requirements in projects involving development of safety critical products in domains like defense, medical and aerospace. Developers can manage requirements in a hierarchical way, linking requirements between themselves and with test cases, identifying dependencies after changing a requirement, creating baselines, approving changes according to roles and exchanging information with partners as excel and reqif files. The scope of the tool is based on the feature set of IBM DOORS.

TRAQS also includes an AI agent that analyses documents and requirements, classifying documents by domains and assessing quality of requirements. Requirements will be rewritten following the rules of selected domain. AI agent also suggests links between requirements and test cases, uses a locally hosted language model and all suggestions should be approved by a user.

1.2 Constraints

TRAQS is a software product, so some constraints apply in an indirect form. Each paragraph below states the constraint and how the project responds to it.

Economic : There is nothing except for the computers of the team and university resources that will be used to finance this project. All parts of the software stack consist of freely available software (React frontend, relational database, open weight language model).

Environmental : Operating a language model requires energy consumption. TRAQS does not train any models; it only performs inference using an already existing model, quantized and smaller in size, only when the user initiates an AI action. Rule-based checks are taken care of in the scenario when there is no requirement of a model.

Social : The stakeholders in this use-case include requirement engineers, reviewers, and suppliers who collaborate from different companies; thus, there should be no need for software installation for communication purposes. The user interface will be provided in both Turkish and English. The purpose of AI is to assist engineers, not replace their decisions; the agent suggests, but people decide.

Political : Data related to defense and aerospace applications is frequently controlled or classified, and the same goes for medical data. In this case, the AI agent will be run locally on the UR-15 model exclusively, and all the data will stay within the organization. In case there is any personal data, it has to be processed according to the KVKK law and also the GDPR for European clients. TRAQS implements DOORS features just as an idea but does not use any IBM code and naming convention.

Ethical : The AI's recommendations may be incorrect or biased, and an unrecognized incorrect rewriting of the requirement might get into a safety-critical product. Every AI output, therefore, is a recommendation that is not accepted by a user, and all actions taken by an AI are logged using the name of the model with a reason and confidence rating (UR-13, UR-15).

Health and safety : The TRAQS system itself doesn't control any physical system, but those products which are associated with TRAQS might, and poor requirement management could result in some adverse effects in the future. TRAQS lowers these risks by means of suspect links, impact analysis, append-only audit log, immutable baselines and locked objects (UR-04, UR-05, UR-06). It's a supportive tool, and it is not certified in accordance with the DO-178C or IEC 62304 standards; the responsibility lies on the user's side.

Manufacturability: For software products, this implies that they can be built, deployed and replicated consistently. TRAQS shall begin its operation from a single command within a containerized image and shall run on Windows and Linux hosts with an inbuilt database, hence requiring no client setup. The AI service is a completely different component with a customizable model endpoint, and thus will be independently built, tested and deployed. All third-party libraries shall have licenses that fit within the academic and open-source environment.

Sustainability : The tool must remain valuable even after the course is completed. Domain profiles and linking rules are configuration files rather than code, meaning that they can easily be modified or extended without altering the program itself. The REST API is versioned and documented, and data may be imported/exported using open formats (Excel/CSV, ReqIF). The language model may also be replaced without modification to the rest of the toolset.

1.3 Professional and Ethical Issues

The group complies with the ACM Code of Ethics and Professional Conduct [1], Software Engineering Code of Ethics [2] and the IEEE Code of Ethics [3]. Additional ethical considerations from computer ethics literature [4] is that any systems automating judgment would pose problems of accountability and transparency. The main ethical considerations for TRAQS are as follows:

Public interest and safety: The code of ethics considers the safety, health, and welfare of the public to be of utmost importance as compared to any other interest [2][3], and calls upon the professionals to prevent damage [1]. TRAQS will be applied to those products in which an omitted requirement or an incorrect modification may pose risks to people. Therefore, it is imperative on the part of the team to consider traceability capabilities (suspect links, impact analysis, audit log, baselines) as a critical component of the system.

Honesty about what the AI can do: The AI agent can produce fluent but wrong text. The team will describe its accuracy honestly, report the results of its evaluation on a labeled reference set, and show reasons and confidence with every suggestion, in line with the duty to be honest and trustworthy [1] and to make only well-founded claims [3]. The rule-based fallback is documented as a lower-capability mode, not as equivalent to the model.

Responsibility and human oversight : The responsibility for the requirement must always be retained by the individual. The agent does not modify a requirement on its own; its output remains in the suggestion queue until the user accepts, modifies, or rejects it, and all the actions performed by the AI are logged with the name of the model, and the audit log tracks who has made what change (UR-05, UR-15).

Confidentiality and privacy : Project files can also be confidential or classified and may even contain personally identifiable information. According to the ethical code, there is a need for privacy and confidentiality [1]. Hence, in TRAQS, role-based access control and project isolation (UR-11) and encrypted communication using TLS without sending any data to any third party (UR-15) have been implemented. Sample projects for the demonstration will use fabricated or public information only.

Intellectual property and licensing: Only those software, modeling tools, and data sets, which have licenses permitting their usage in the manner intended will be used. All outside work will be properly credited, and there will be no copying of any of the code, documentation, or trademarks from the IBM DOORS or other products [1]. The domain rules will come from public standard descriptions and will be duly acknowledged.

Quality and professional competence: The codes require excellent performance and objective assessment of risks and limitations [1][2]. The group will perform testing on requirement-based functionalities, note down the limitations in the final report, and request the supervisor to validate the domain guidelines since the team members lack domain expertise.

Teamwork and fair contribution: Tasks are separated according to their named owners and estimates, progress is done in public, and contributions from each team member are accurately measured. The use of any AI technologies for coding and writing will be revealed according to the course guidelines.

2. Requirements

The following list contains the first requirements of TRAQS, as perceived by the users. These requirements are based on the features offered by IBM DOORS, but supplemented with the addition of the AI agent and the change management process that was required by the supervisor. Prioritization of requirements follows MoSCoW convention (Must, Should, Could). Each requirement ends with the main functions of the system that result from it.

ID	Initial requirement	Priority	Main system functions
UR-01	The user shall be able to capture and organize requirements in hierarchical modules.	Must	Projects, modules and ordered object hierarchy; unique, never-reused IDs
UR-02	The user shall be able to define custom attributes and filtered views of requirements.	Must	User-defined typed attributes; saved views with filters, columns and sorting
UR-03	The user shall be able to link requirements to other requirements, design items and test cases.	Must	Typed links (Satisfies, Derives, Verifies, Refers); link rule validation; traceability matrix with upstream/downstream view
UR-04	The user shall be able to see which items are affected when a requirement changes.	Must	Suspect links on change; impact analysis
UR-05	The user shall be able to propose, review and approve changes through a change management workflow.	Must	Change request lifecycle; role-based approval; locking of approved objects; audit log
UR-06	The user shall be able to freeze, compare and restore baselines and see the full change history.	Must	Immutable baselines with safe restore; baseline comparison; per-object change history
UR-07	The user shall be able to use the system from a web browser and discuss requirements with other team members.	Must	Web application; threaded comments; notifications; concurrent editing control; stakeholder validation
UR-08	The user shall be able to exchange requirements with suppliers and	Must	Excel/CSV import and export; ReqIF import and export (subset)

	partners in Excel and ReqIF formats.		
UR-09	The user shall be able to link requirements to test cases and record test results.	Must	Test case management; verification status derived from test results
UR-10	Other engineering tools shall be able to read and write project data through an open interface (REST API).	Should	Documented, versioned REST API
UR-11	The user shall only access the projects and functions permitted by the assigned role.	Must	Authentication and role-based permissions; per-project data isolation; user and group management; TLS and CSRF protection
UR-12	The AI agent shall separate requirements by domain (defense, medical) and by type.	Must	Domain profiles; classification by domain, type and criticality
UR-13	The AI agent shall check the quality of each requirement.	Must	Quality score (0 to 100) with reasons, from rules and an LLM review
UR-14	The AI agent shall revise requirements to match the rules of the selected domain.	Must	Domain-specific rewrite proposals with rationale, standard reference and diff
UR-15	The AI agent shall draft requirements from documents and suggest traceability links. It shall run on a local language model, and its suggestions shall require user approval.	Must	Document ingestion and requirement drafting; link suggestions with confidence; local LLM; suggestion queue (accept, edit, reject) and AI action log
UR-16	Once the requirements are linked top-down, the system shall analyze the whole chain and report the results.	Must	Chain analysis (coverage, orphans, suspect links, quality, domain compliance); dashboard and exportable report
UR-17	The team shall be able to demonstrate the application with realistic sample projects.	Must	Sample defense and medical projects; scripted end-to-end demo scenario

3. References

1. Association for Computing Machinery, "ACM Code of Ethics and Professional Conduct," 2018. [Online]. Available: https://www.acm.org/code-of-ethics
2. ACM/IEEE Computer Society Joint Task Force on Software Engineering Ethics and Professional Practices, "Software Engineering Code of Ethics and Professional Practice." [Online]. Available: https://ethics.acm.org/code-of-ethics/software-engineering-code/
3. IEEE, "IEEE Code of Ethics," IEEE Policies, Section 7.8. [Online]. Available: https://www.ieee.org/about/corporate/governance/p7-8.html
4. T. W. Bynum, "Computer and Information Ethics," in *The Stanford Encyclopedia of Philosophy*. [Online]. Available: https://plato.stanford.edu/entries/ethics-computer/

Other documents mentioned in the text: Protection of Personal Data (KVKK, Turkey); Regulation (EU) 2016/679 (GDPR); RTCA DO-178C; IEC 62304; the OMG Requirements Interchange Format (ReqIF); IBM DOORS product documentation.